Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20160118-004-DS-2-RV-102	Issued	2016-02-12

UT Thickness Data Sheet - 2-RV-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 2-RV-102 located within the ATU-100. The scope of work involves the recording of wall thickness to monitor asset integrity and determine remaining life based on established site retirement limits.

Marcus Chen, acting as the lead UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and is responsible for the accuracy of the recorded data.

Technical Basis and Procedures

All thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 2. This procedure includes specific mandates for insulation removal and reinstatement, inspection-port requirements, and profile radiography to support the CUI program.

The inspection methodology and instrument calibration were conducted in accordance with ASTM E797.

Line Specifications

The subject line, 2-RV-102, is a regenerator overhead circuit constructed from A106 Gr.B. The line has a nominal outer diameter of 60.3 mm and a nominal wall thickness of 5.54 mm.

The design parameters for this line are 345 kPa(g) and 121 °C. The minimum allowable thickness (t-min) is 2.79 mm.

Inspection Context

The data recorded herein is part of the broader survey detailed in LNDE-RPT-20160118-004-R0. This full-unit survey covers the ATU-100, including the 3-year thickness interval required for the regenerator overhead and rich amine circuits.

The current assessment ensures compliance with the 3-year interval for the regenerator overhead circuit as specified in the site integrity rules.

Thickness readings, measured 2016-02-12

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2011-02-18	6.39	6.17	0.048	0.044	>50	2021-02-12	
6-SG-101-02	7.11	3.56	2011-02-18	6.14	5.76	0.065	0.076	28.8	2021-02-12	
6-SG-101-03	7.11	3.56	2011-02-18	6.17	5.87	0.061	0.060	38.1	2021-02-12	
6-SG-101-04	7.11	3.56	2011-02-18	6.05	5.58	0.076	0.094	21.4	2021-02-12	
6-SG-101-05	7.11	3.56	2011-02-18	5.87	5.38	0.088	0.098	18.5	2021-02-12	
6-SG-101-06	7.11	3.56	2011-02-18	6.27	5.88	0.060	0.078	29.6	2021-02-12	
6-SG-101-08	7.11	3.56	2011-02-18	6.33	6.00	0.066	0.066	36.8	2021-02-12	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2011-02-18	5.75	5.42	0.086	0.066	21.6	2021-02-12	
6-SG-102-02	7.11	3.56	2011-02-18	5.80	5.43	0.084	0.074	22.2	2021-02-12	
6-SG-102-03	7.11	3.56	2011-02-18	5.80	5.22	0.091	0.116	14.3	2021-02-12	
6-SG-102-04	7.11	3.56	2011-02-18	5.73	5.18	0.091	0.110	14.7	2021-02-12	
6-SG-102-05	7.11	3.56	2011-02-18	6.46	6.19	0.047	0.054	48.5	2021-02-12	
6-SG-102-06	7.11	3.56	2011-02-18	6.41	6.06	0.070	0.070	35.6	2021-02-12	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2006-02-24	7.89	7.76	0.022	0.013	>50	2026-02-12	
8-TG-101-02	8.18	4.06	2006-02-24	7.75	7.46	0.037	0.029	>50	2026-02-12	
8-TG-101-03	8.18	4.06	2006-02-24	7.79	7.54	0.035	0.025	>50	2026-02-12	
8-TG-101-04	8.18	4.06	2006-02-24	7.76	7.51	0.030	0.025	>50	2026-02-12	
8-TG-101-05	8.18	4.06	2006-02-24	7.95	7.73	0.025	0.022	>50	2026-02-12	
8-TG-101-07	8.18	4.06	—	—	7.39	—	—	—	2026-02-12	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2006-02-24	5.51	5.02	0.049	0.049	37.4	2026-02-12	
4-LA-101-02	6.02	3.18	2006-02-24	5.76	5.45	0.028	0.031	>50	2026-02-12	
4-LA-101-03	6.02	3.18	2006-02-24	5.76	5.45	0.026	0.031	>50	2026-02-12	
4-LA-101-04	6.02	3.18	2006-02-24	5.78	5.48	0.030	0.030	>50	2026-02-12	
4-LA-101-05	6.02	3.18	2006-02-24	5.81	5.54	0.024	0.027	>50	2026-02-12	
4-LA-101-06	6.02	3.18	2006-02-24	5.64	5.30	0.038	0.034	>50	2026-02-12	
4-LA-101-07	6.02	3.18	2006-02-24	5.57	4.87	0.056	0.070	24.1	2026-02-12	
4-LA-101-09	6.02	3.18	—	—	5.00	—	—	—	2026-02-12	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2006-02-24	5.63	5.25	0.039	0.038	>50	2026-02-12	
4-LA-102-02	6.02	3.18	2006-02-24	5.67	5.36	0.036	0.031	>50	2026-02-12	
4-LA-102-03	6.02	3.18	2006-02-24	5.55	5.28	0.040	0.027	>50	2026-02-12	
4-LA-102-04	6.02	3.18	2006-02-24	5.34	4.90	0.055	0.044	31.5	2026-02-12	
4-LA-102-05	6.02	3.18	2006-02-24	5.62	5.34	0.033	0.028	>50	2026-02-12	
4-LA-102-06	6.02	3.18	2006-02-24	5.83	5.60	0.020	0.023	>50	2026-02-12	
4-LA-102-07	6.02	3.18	2006-02-24	5.63	5.40	0.030	0.023	>50	2026-02-12	
4-LA-102-08	6.02	3.18	—	—	5.70	—	—	—	2026-02-12	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2006-02-24	6.75	6.44	0.029	0.031	>50	2026-02-12	
6-LA-103-02	7.11	3.56	2006-02-24	6.87	6.53	0.033	0.034	>50	2026-02-12	
6-LA-103-03	7.11	3.56	2006-02-24	6.65	6.28	0.040	0.037	>50	2026-02-12	
6-LA-103-04	7.11	3.56	2006-02-24	6.52	6.15	0.051	0.037	>50	2026-02-12	
6-LA-103-05	7.11	3.56	2006-02-24	6.49	5.98	0.061	0.051	39.9	2026-02-12	
6-LA-103-06	7.11	3.56	2006-02-24	6.42	5.82	0.067	0.060	33.6	2026-02-12	
6-LA-103-07	7.11	3.56	2006-02-24	6.76	6.12	0.053	0.064	39.9	2026-02-12	
6-LA-103-08	7.11	3.56	2006-02-24	6.79	6.49	0.028	0.030	>50	2026-02-12	
6-LA-103-10	7.11	3.56	—	—	6.30	—	—	—	2026-02-12	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2014-09-19	8.07	7.83	0.035	0.098	40.9	2019-02-12	
4-RA-101-02	8.56	3.81	2014-09-19	8.02	7.94	0.030	0.052	>50	2019-02-12	
4-RA-101-03	8.56	3.81	2014-09-19	7.88	7.65	0.048	0.066	>50	2019-02-12	
4-RA-101-04	8.56	3.81	2014-09-19	8.18	8.00	0.032	0.042	>50	2019-02-12	
4-RA-101-05	8.56	3.81	2014-09-19	7.98	7.82	0.039	0.068	>50	2019-02-12	
4-RA-101-06	8.56	3.81	2014-09-19	7.91	7.87	0.029	0.029	>50	2019-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-07	8.56	3.81	2014-09-19	8.22	8.14	0.023	0.032	>50	2019-02-12	
4-RA-101-08	8.56	3.81	2014-09-19	7.71	7.31	0.069	0.142	24.6	2019-02-12	
4-RA-101-09	8.56	3.81	2014-09-19	8.27	8.13	0.022	0.038	>50	2019-02-12	
4-RA-101-10	8.56	3.81	2014-09-19	8.21	8.02	0.033	0.052	>50	2019-02-12	
4-RA-101-11	8.56	3.81	2014-09-19	7.88	7.60	0.054	0.088	42.9	2019-02-12	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2014-09-19	8.25	8.02	0.026	0.062	>50	2019-02-12	
4-RA-102-02	8.56	3.81	2014-09-19	7.89	7.64	0.047	0.070	>50	2019-02-12	
4-RA-102-03	8.56	3.81	2014-09-19	7.95	7.55	0.051	0.098	38.0	2019-02-12	
4-RA-102-04	8.56	3.81	2014-09-19	7.93	7.72	0.041	0.048	>50	2019-02-12	
4-RA-102-05	8.56	3.81	2014-09-19	7.84	7.67	0.044	0.080	48.1	2019-02-12	
4-RA-102-06	8.56	3.81	2014-09-19	7.95	7.73	0.039	0.056	>50	2019-02-12	
4-RA-102-07	8.56	3.81	2014-09-19	7.98	7.78	0.040	0.064	>50	2019-02-12	
4-RA-102-08	8.56	3.81	2014-09-19	7.58	7.35	0.057	0.106	33.3	2019-02-12	
4-RA-102-09	8.56	3.81	2014-09-19	7.84	7.54	0.051	0.088	42.2	2019-02-12	
4-RA-102-11	8.56	3.81	2014-09-19	8.12	7.96	0.046	0.046	>50	2019-02-12	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2014-09-19	10.62	10.29	0.036	0.036	>50	2019-02-12	
6-RA-103-02	10.97	4.57	2014-09-19	10.06	9.69	0.070	0.108	47.2	2019-02-12	
6-RA-103-03	10.97	4.57	2014-09-19	10.39	10.06	0.047	0.072	>50	2019-02-12	
6-RA-103-04	10.97	4.57	2014-09-19	9.79	9.31	0.084	0.132	35.8	2019-02-12	
6-RA-103-05	10.97	4.57	2014-09-19	9.81	9.48	0.074	0.124	39.5	2019-02-12	
6-RA-103-06	10.97	4.57	2014-09-19	10.43	10.32	0.035	0.048	>50	2019-02-12	
6-RA-103-07	10.97	4.57	2014-09-19	10.37	10.20	0.040	0.070	>50	2019-02-12	
6-RA-103-08	10.97	4.57	2014-09-19	10.49	10.33	0.032	0.022	>50	2019-02-12	
6-RA-103-09	10.97	4.57	2014-09-19	10.39	10.33	0.034	0.046	>50	2019-02-12	
6-RA-103-10	10.97	4.57	2014-09-19	9.93	9.49	0.077	0.112	43.8	2019-02-12	
6-RA-103-11	10.97	4.57	2014-09-19	10.10	9.77	0.236	0.236	22.0	2019-02-12	
6-RA-103-13	10.97	4.57	2014-09-19	10.38	10.15	0.060	0.060	>50	2019-02-12	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2013-09-09	6.49	6.34	0.063	0.081	34.2	2019-02-12	
3-RV-101-02	7.62	3.56	2013-09-09	6.44	6.31	0.063	0.052	43.5	2019-02-12	
3-RV-101-03	7.62	3.56	2013-09-09	5.80	5.63	0.095	0.098	21.2	2019-02-12	
3-RV-101-04	7.62	3.56	2013-09-09	6.13	5.88	0.086	0.102	22.8	2019-02-12	
3-RV-101-05	7.62	3.56	2013-09-09	6.48	6.36	0.066	0.050	42.6	2019-02-12	
3-RV-101-06	7.62	3.56	2013-09-09	6.11	5.96	0.082	0.087	27.7	2019-02-12	
3-RV-101-07	7.62	3.56	2013-09-09	5.73	5.54	0.122	0.122	16.2	2019-02-12	
3-RV-101-08	7.62	3.56	2013-09-09	6.36	6.31	0.074	0.074	37.2	2019-02-12	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2013-09-09	4.49	4.35	0.059	0.037	26.4	2019-02-12	
2-RV-102-02	5.54	2.79	2013-09-09	4.29	4.01	0.079	0.109	11.2	2019-02-12	
2-RV-102-03	5.54	2.79	2013-09-09	4.68	4.64	0.042	0.037	43.9	2019-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-04	5.54	2.79	2013-09-09	4.81	4.73	0.043	0.020	45.5	2019-02-12	
2-RV-102-05	5.54	2.79	2013-09-09	4.96	4.97	0.013	0.013	>50	2019-02-12	
2-RV-102-06	5.54	2.79	2013-09-09	4.55	4.27	0.102	0.102	14.6	2019-02-12	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2013-12-09	5.37	5.31	0.009	0.017	>50	2026-02-12	
3-RF-101-02	5.49	2.41	2013-12-09	5.40	5.34	0.006	0.003	>50	2026-02-12	
3-RF-101-03	5.49	2.41	2013-12-09	5.37	5.33	0.006	0.012	>50	2026-02-12	
3-RF-101-04	5.49	2.41	2013-12-09	5.41	5.35	0.006	-0.002	>50	2026-02-12	
3-RF-101-05	5.49	2.41	2013-12-09	5.32	5.33	0.006	0.007	>50	2026-02-12	
3-RF-101-06	5.49	2.41	2013-12-09	5.17	5.23	0.015	0.017	>50	2026-02-12	
3-RF-101-07	5.49	2.41	2013-12-09	5.48	5.37	0.003	-0.002	>50	2026-02-12	
3-RF-101-09	5.49	2.41	2013-12-09	5.26	5.19	-0.029	-0.029	>50	2026-02-12	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2013-12-09	3.60	3.76	0.008	0.009	>50	2026-02-12	
2-RF-102-02	3.91	2.16	2013-12-09	3.70	3.64	0.012	0.024	>50	2026-02-12	
2-RF-102-03	3.91	2.16	2013-12-09	3.79	3.75	0.009	0.005	>50	2026-02-12	
2-RF-102-04	3.91	2.16	2013-12-09	3.82	3.38	0.025	0.043	28.3	2026-02-12	
2-RF-102-05	3.91	2.16	2013-12-09	3.67	3.49	0.082	0.082	16.1	2024-03-06	
2-RF-102-06	3.91	2.16	2013-12-09	3.67	3.47	0.099	0.099	13.2	2022-09-26	
2-RF-102-07	3.91	2.16	2013-12-09	3.72	3.57	0.020	0.028	>50	2026-02-12	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2006-02-24	5.75	5.31	0.038	0.044	48.2	2026-02-12	
4-LA-104-02	6.02	3.18	2006-02-24	5.56	5.00	0.052	0.056	32.4	2026-02-12	
4-LA-104-03	6.02	3.18	2006-02-24	5.74	5.42	0.027	0.032	>50	2026-02-12	
4-LA-104-04	6.02	3.18	2006-02-24	5.86	5.63	0.022	0.023	>50	2026-02-12	
4-LA-104-05	6.02	3.18	2006-02-24	5.51	4.88	0.058	0.063	26.9	2026-02-12	
4-LA-104-06	6.02	3.18	2006-02-24	5.83	5.55	0.028	0.028	>50	2026-02-12	
4-LA-104-07	6.02	3.18	2006-02-24	5.62	5.28	0.037	0.034	>50	2026-02-12	
4-LA-104-08	6.02	3.18	2006-02-24	5.50	5.11	0.042	0.039	46.3	2026-02-12	
4-LA-104-10	6.02	3.18	—	—	4.84	—	—	—	2026-02-12	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2014-09-19	7.84	7.71	0.040	0.048	>50	2019-02-12	
4-RA-104-02	8.56	3.81	2014-09-19	8.09	7.85	0.035	0.058	>50	2019-02-12	
4-RA-104-03	8.56	3.81	2014-09-19	8.46	8.28	0.057	0.056	>50	2019-02-12	
4-RA-104-04	8.56	3.81	2014-09-19	8.49	8.18	0.063	0.084	>50	2019-02-12	
4-RA-104-05	8.56	3.81	2014-09-19	8.37	8.16	0.050	0.082	>50	2019-02-12	
4-RA-104-06	8.56	3.81	2014-09-19	7.91	7.61	0.049	0.074	>50	2019-02-12	
4-RA-104-07	8.56	3.81	2014-09-19	7.61	7.32	0.064	0.092	38.0	2019-02-12	
4-RA-104-08	8.56	3.81	2014-09-19	8.00	7.81	0.074	0.074	>50	2019-02-12	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2006-02-24	3.64	3.61	0.016	0.003	>50	2026-02-12	
2-ST-101-02	3.91	2.29	2006-02-24	3.67	3.55	0.018	0.012	>50	2026-02-12	
2-ST-101-03	3.91	2.29	2006-02-24	3.82	3.55	0.020	0.027	46.5	2026-02-12	
2-ST-101-04	3.91	2.29	2006-02-24	3.77	3.68	0.007	0.009	>50	2026-02-12	
2-ST-101-05	3.91	2.29	2006-02-24	3.72	3.50	0.021	0.022	>50	2026-02-12	
2-ST-101-06	3.91	2.29	—	—	3.63	—	—	—	2026-02-12	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2006-02-24	5.13	5.03	0.020	0.010	>50	2026-02-12	
3-CW-101-02	5.49	2.79	2006-02-24	5.20	5.01	0.020	0.019	>50	2026-02-12	
3-CW-101-03	5.49	2.79	2006-02-24	5.08	4.93	0.030	0.015	>50	2026-02-12	
3-CW-101-04	5.49	2.79	2006-02-24	5.38	5.26	0.010	0.012	>50	2026-02-12	
3-CW-101-05	5.49	2.79	2006-02-24	5.33	5.35	0.007	-0.002	>50	2026-02-12	
3-CW-101-06	5.49	2.79	—	—	5.08	—	—	—	2026-02-12	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2006-02-24	5.20	5.06	0.022	0.014	>50	2026-02-12	
3-CW-102-02	5.49	2.79	2006-02-24	5.02	4.37	0.057	0.065	24.2	2026-02-12	
3-CW-102-03	5.49	2.79	2006-02-24	5.07	4.54	0.050	0.053	32.9	2026-02-12	
3-CW-102-04	5.49	2.79	2006-02-24	5.03	4.69	0.041	0.034	46.2	2026-02-12	
3-CW-102-05	5.49	2.79	—	—	4.66	—	—	—	2026-02-12	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2016-02-12	_____